Layer 1: Client Presentation Layer (Next.js 16.4.2 React UI)

Continuous Monitoring & Drift Radar

Investor Dashboard & NAV View

Copilot Chat & Fiduciary Stock-Why

Portfolio Wizard & Risk Profiler

Rebalance Execution & Order Preview

Demat Statement Diagnostics (HHI)

Layer 2: API Gateway & Middleware Layer (FastAPI 0.115)

In-Memory Sliding-Window Rate Limiter

FastAPI Core Application (backend/main.py)

Modular REST Routers (9 Routers, 82 Endpoints)

JWT Authentication & RBAC (backend/auth/)

Strict HTTP Security Headers (CSP, HSTS, XFO)

CORS Policy Validation (localhost:3000)

Layer 3: Orchestration & State Management Layer (LangGraph)

Conditional Routing & Fiduciary Gate Logic

StateGraph Orchestrator (backend/agent/graph.py)

PortfolioAgentState TypedDict (backend/agent/state.py)

Full Step-by-Step Telemetry & LLM Audit Logging

Layer 4: Agentic AI & Intelligence Layer (13 Specialized Nodes)

Macro Analyst Node (RBI & Inflation)

Manager Node (Task Decomposition)

News & Corporate Filings Analyst Node

Fiduciary Moderator Node (Two-Level Explanations)

Lead Strategist Node (Tactical Candidate Weights)

ACID PostgreSQL Persistence Node

Suitability & Feasibility Node (SEBI Gate)

Deterministic SLSQP Optimizer Node

Independent Risk Critic Node (Mandate Audit)

Market Analyst Node (NIFTY 50 50/200 DMA)

Fundamental Analyst Node (P/E, ROE, Factors)

Strategist Bounded Rework Node (Max 2 Loops)

Critic Final Validation Node (Constraint Audit)

Layer 6: Operational Services & Scheduler Layer

Demat Diagnosis Service (HHI Concentration & Overlap)

Background Monitoring Scheduler (300s, 09:15-15:30 IST)

Security Universe Sync Service (NSE/BSE Ingestion)

Notification Service (In-App Unread Alerts)

Stock-Why Service (Institutional Citations & Proofs)

Rebalancing Service (Trade Preview & Invariant Checks)

Layer 5: Quantitative & Mathematical Engine Layer (SciPy & NumPy)

Asset & Sector Allocation Drift Engine (Threshold: 5%/8%)

Deterministic SLSQP Optimizer (Markowitz Quadratic Utility)

Risk Engine (Sharpe, Sortino, Beta, Max Drawdown)

Whole-Share Integer Allocation & Cash Conservation Engine

Statutory Indian Equity Transaction Costs Engine (STT, GST)

Multi-Factor Scoring & Winsorized Z-Score Normalizer

Layer 7: Data Access & Persistence Layer (PostgreSQL 16 Master)

Security Master (2,639 Securities, 2,715 Listings)

SQLAlchemy 2.0 ORM & Connection Pooling

PostgreSQL 16 Database (majorproject) 62 Domain Tables + alembic_version

Portfolio Store (Versions, Decisions, Trades)

Telemetry & System Audit Logs

Layer 8: External Integrations & Market Data Layer

Yahoo Finance API (yfinance Live Quotes & History)

NSE India Archives (Official EQUITY_L Master)

Google Gemini API (gemini-3.8-flash / Local Fallback)

BSE India Corporate Directory (ISIN Cross-Reference)

Reserve Bank of India Bulletins (Policy Repo Rate)

NewsAPI & RSS Financial News Feeds